

E-Commerce Customer Insights and sales Project

What is this project?

Olist is a Brazilian e-commerce company with 100,000+ orders data. I took their raw messy data, cleaned it, analyzed it using SQL, and made visualizations to find out what is working and what is not in their business.

What I Did

Data Cleaning — 9 CSV files, 1M+ rows. Fixed wrong data types, handled nulls, removed useless rows, fixed column name typos. Saved all 9 cleaned files separately.

SQL Analysis — Loaded cleaned data into SQLite and wrote 9 KPI queries to calculate revenue, average order value, repeat customer rate, late delivery rate and more.

Data Visualization — Made 7 Seaborn charts to find patterns in data that numbers alone cannot show.

What I Found

- **Total Revenue** → R\$ 16,008,872 in 2 years

- **20x Growth** — monthly revenue went from R\$50K in 2016 to R\$1M+ by 2018
- **Black Friday Spike** — November 2017 was the single highest revenue month
- **Health & Beauty** is the top earning category at R\$1.25M
- Late Deliveries are directly causing 1 star reviews — biggest operational problem
- **40% orders from São Paulo** — rest of Brazil is barely buying
- **Repeat rate is very low** — most customers buy once and never return

5 Business Recommendations

- Focus more on Health & Beauty category
- Fix late deliveries to improve review scores
- Start a loyalty program to bring customers back
- Target other states beyond São Paulo
- Prepare inventory and logistics for Black Friday every year

Tech Stack

Python, Pandas, SQL, Seaborn, Matplotlib, Power BI

Dataset

Olist Brazilian E-Commerce — Kaggle — 100,000+ orders, 9 tables, 2016 to 2018